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HALI-net: An explainable hybrid deep learning model with attention and texture fusion for lung cancer detection in CT images

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ABSTRACT

Background: Lung cancer is among the leading causes of cancer-related mortality worldwide, and early detection from CT images is critical for improving patient outcomes. Manual interpretation of CT scans is time-consuming, subjective, and susceptible to inter-observer variability, necessitating automated and reliable computer-aided diagnosis systems.

Method: This study proposes HALI-Net, a hybrid attention-based deep learning framework for multiclass lung cancer classification from CT scan images. The framework integrates adaptive Gaussian-based preprocessing, handcrafted texture feature extraction using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP), and deep feature extraction via InceptionV3 enhanced with a Multi-Scale Channel Attention (MSCA) mechanism. A hybrid feature fusion strategy combines deep and handcrafted representations, which are subsequently classified by an MLP classifier. Gradient-weighted Class Activation Mapping (Grad-CAM) is incorporated to provide visual interpretability. The model is evaluated on the IQ-OTH/NCCD lung cancer dataset comprising three diagnostic classes: Benign, Malignant, and Normal.

Results: HALI-Net achieves a classification accuracy of 98.55%, with strong precision, recall, F1- score, and a ROC-AUC of 99.90%. Ablation studies confirm the individual contribution of each proposed component, and Grad-CAM visualizations demonstrate consistent focus on clinically relevant pulmonary regions across all diagnostic categories.

Conclusion: The proposed HALI-Net framework delivers an accurate, computationally efficient, and clinically interpretable solution for automated lung cancer detection from CT images. Its hybrid architecture and explainability mechanisms make it well-suited for integration into computer-aided diagnostic workflows in clinical environments.

1. Introduction

Lung cancer continues to rank among the deadliest forms of cancer, contributing to a striking share of global cancer mortality each year. One reason is timing. Patients diagnosed at an early stage often experience markedly better outcomes, yet early-stage tumors can be subtle and difficult to identify consistently in routine screening workflows. In clinical settings, Computed Tomography (CT) has become the standard imaging technique for lung cancer assessment because it provides detailed cross-sectional views of pulmonary structures, nodules, and surrounding tissues. Even so, interpreting hundreds of CT slices for a single patient is rarely straightforward. Radiologists frequently work under substantial time pressure, and diagnostic decisions may vary between specialists, particularly when lesions are small, irregular, or visually ambiguous. As imaging volumes continue to increase across hospitals and diagnostic centers, these practical constraints have

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intensified interest in automated Computer-Aided Detection (CAD) systems that could support clinicians during screening and diagnostic evaluation [1].

Deep learning methods have attracted considerable attention in medical image analysis over the last decade. Their influence is especially visible in tasks such as lesion detection, organ segmentation, and disease classification. Convolutional Neural Networks (CNNs), in particular, appear well suited for extracting hierarchical image representations directly from raw scans, often reducing the dependence on manually engineered features. Architectures such as InceptionV3 have shown promising performance in several medical imaging studies, partly because the network processes information at multiple receptive-field scales while remaining computationally manageable during transfer learning [2]. Yet high predictive accuracy does not necessarily imply complete feature understanding. In CT-based lung cancer analysis, subtle texture variations inside nodules or surrounding tissues may carry diagnostic value that purely deep feature representations do not always preserve effectively.

For this reason, handcrafted texture descriptors still retain relevance despite the dominance of deep learning approaches. Techniques such as Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP) were originally developed to quantify structural and statistical image properties that human observers often recognize intuitively but struggle to measure consistently. GLCM evaluates relationships between neighboring pixel intensities and can describe characteristics including contrast, homogeneity, energy, and correlation [3]. LBP operates differently. It encodes local texture patterns by comparing a central pixel with its surrounding neighborhood, generating compact representations that are relatively stable under rotational variation. When these descriptors are combined with CNN-derived features, the resulting representation may capture both semantic information and fine-grained tissue texture. This combination could be particularly useful in smaller medical datasets, where deep networks alone sometimes overfit or fail to generalize reliably [4].

Another line of progress has emerged through attention mechanisms. Rather than processing all regions of an image with equal emphasis, attention modules attempt to guide the network toward diagnostically informative areas. In lung CT scans, that distinction matters because suspicious nodules may occupy only a very limited portion of the image while surrounding anatomical structures dominate the visual field. Channel attention mechanisms help the model weigh informative feature maps more strongly, whereas spatial attention highlights relevant image regions directly [5]. Used together, these mechanisms may improve sensitivity to subtle lesions that could otherwise be diluted during feature extraction. The effect is not merely computational; it also aligns more closely with how radiologists visually prioritize suspicious regions during examination.

Despite these advances, one persistent concern remains difficult to ignore: interpretability. Many deep learning systems generate highly accurate predictions while offering limited insight into how those predictions were formed. In medical contexts, that opacity can create hesitation among clinicians, especially when diagnostic decisions influence treatment planning or invasive follow-up procedures [6]. Explainable Artificial Intelligence (XAI) techniques attempt to address this gap by exposing portions of the model's reasoning process. Among these methods, Gradient-weighted Class Activation Mapping (Grad-CAM) has become widely used because it produces visual heatmaps indicating which image regions contributed most strongly to a prediction. Although such explanations do not fully resolve concerns surrounding transparency, they may help clinicians evaluate whether the model is focusing on anatomically meaningful regions rather than irrelevant artifacts [7].

Motivated by these considerations, this study introduces HALI-Net, a Hybrid Attention-based Lung Intelligence Network designed for explainable lung cancer detection from CT imagery. The framework combines several complementary components rather than relying on a single modeling strategy. An adaptive Gaussian preprocessing stage is first employed to reduce image noise while attempting to preserve diagnostically relevant boundaries and tissue structures. Deep semantic features are then extracted using InceptionV3 and refined through a Multi-Scale Channel Attention (MSCA) mechanism. In parallel, handcrafted texture descriptors derived from GLCM and LBP are incorporated to retain local structural characteristics that may not be fully represented within deep feature maps. These heterogeneous features are subsequently fused through a Multi-Layer Perceptron (MLP)-based classifier [8]. To improve interpretability, Grad-CAM visualizations are integrated into the final framework so that prediction-relevant regions can be inspected directly. Experimental evaluation is conducted using the IQ-OTHNCCD lung cancer dataset, which contains three diagnostic categories: Benign, Malignant, and Normal CT cases.

Although individual components such as CNN backbones, attention mechanisms, handcrafted texture descriptors, and Grad-CAM explainability have each appeared in prior works, HALI-Net distinguishes itself through a specific architectural integration that has not been previously explored in this combination. The MSCA module applies sequential channel and spatial attention directly after the mixed7 layer of InceptionV3, a design choice that differs from standard CBAM-style insertion points. The adaptive Gaussian preprocessing dynamically modulates smoothing intensity based on local Laplacian variance rather than applying uniform filtering. The hybrid fusion of 58-dimensional handcrafted descriptors with 128-dimensional MSCA-refined deep features into a unified 186-dimensional representation fed to an MLP constitutes a structurally distinct pipeline from methods that concatenate features without attention-guided refinement. Together, these decisions reflect a framework engineered for both diagnostic accuracy and clinical deployment feasibility.

The main contributions of this work can be summarized as follows:

- A hybrid classification architecture, HALI-Net, is developed by combining CNN-derived semantic representations with handcrafted texture descriptors extracted from CT images.
- An adaptive Gaussian preprocessing strategy is introduced to suppress imaging noise while maintaining structural details that may carry diagnostic significance.
- A Multi-Scale Channel Attention (MSCA) module is incorporated to encourage the network to focus more selectively on informative lung regions and feature channels.
- A feature-level fusion strategy integrating InceptionV3, GLCM, and LBP representations is designed to strengthen discriminatory performance across multiple lung cancer categories.
- A Grad-CAM-based explainability component is included to provide visual interpretation of model predictions and support greater transparency during clinical assessment.

The remainder of this paper is structured as follows. Section II surveys prior studies related to lung cancer detection, texture-based analysis, attention mechanisms, and explainable AI techniques in medical imaging. Section III presents the proposed HALI-Net framework, including preprocessing, feature extraction, attention integration, fusion methodology, and interpretability design. Section IV outlines the experimental configuration, dataset characteristics, and evaluation metrics. Section V discusses the obtained results and comparative analysis. Finally, Section VI concludes the study and identifies several directions that may warrant further investigation.

1.1. Related work & motivation

Recent work on lung cancer detection from CT imagery increasingly points toward hybrid learning frameworks rather than single-model solutions. Researchers seem to be moving away from relying solely on conventional CNN pipelines and are instead combining convolutional networks with Vision Transformers (ViTs), attention modules,

handcrafted texture descriptors, and explainable AI techniques. The rationale is fairly intuitive. CNNs tend to capture localized structural patterns effectively, whereas Transformer-based models may better preserve broader contextual relationships across lung regions and surrounding tissue structures.

Among the more notable contributions, Ozdemir proposed a hybrid CNN–Vision Transformer architecture that integrated InceptionNeXt with both grid and block attention mechanisms for multiclass lung cancer classification. Their experiments on the Chest CT and IQ-OTH/NCCD datasets produced remarkably high results—99.54% accuracy on IQ-OTH/NCCD and 98.41% on the Chest CT dataset [9]. What makes the study particularly interesting is not only the reported accuracy, but also the attempt to combine transfer learning with targeted preprocessing strategies to stabilize feature extraction across varying CT slice qualities.

A somewhat different direction was taken by Haque and colleagues, who designed an end-to-end concatenated CNN attention model incorporating dual CNN branches, multilayer perceptrons, and multi-head attention. Their framework also relied heavily on interpretability tools such as Grad- CAM and SHAP, suggesting an awareness that clinical adoption may depend as much on transparency as on predictive performance. The model achieved 99.54% accuracy alongside a recall value of 99.95%, which implies that false negatives were relatively limited in their experimental setting [10].

Several studies attempted to bridge handcrafted radiological descriptors with deep feature learning rather than replacing one with the other. For example, Badawy developed a system combining GLCM-style handcrafted features with CNN, CoAtNet, and EfficientNet architectures using more than 30,000 CT images from the BOWL2017 dataset. Their best-performing configuration reached approximately 98% accuracy and slightly higher precision values [11]. That finding is worth noting because handcrafted texture representations—once considered somewhat outdated in deep learning literature—still appear capable of capturing subtle tissue irregularities that purely end-to- end networks may overlook.

Interpretability has also become a recurring theme across recent literature. Rao introduced LungDxFormer, a CNN–Transformer framework equipped with Dynamic Spatial Attention for multiclass lung nodule analysis. Using Grad-CAM visualizations on the LIDC-IDRI dataset, the study reported 97.35% classification accuracy while also showing where the model concentrated during inference [12]. In clinical imaging tasks, that spatial transparency matters. A prediction without anatomical justification can be difficult for radiologists to trust, especially in borderline benign cases where visual boundaries are often ambiguous.

Other researchers focused more aggressively on optimization and ensemble learning strategies. Al proposed the SE-FusionEAOO framework, which combined squeeze-and-excitation blocks, SMOTE balancing, and evolutionary optimization through EAOO-GA. Their system achieved 99.40% accuracy on IQ-OTH/NCCD and maintained strong performance during external validation on LIDC-IDRI [13]. Meanwhile, Sudeshna explored GAN-assisted augmentation through LungGANDetectAI, integrating ResNet50, SCAM attention, and Grad-CAM guidance. The approach reached roughly 98.4% accuracy, suggesting that synthetic augmentation might help compensate for limited annotated medical datasets [14].

Not every framework pursued extreme classification scores. Some studies instead concentrated on balancing segmentation, temporal modeling, and computational feasibility. Khurram combined CNNs, LSTM units, and transformer self-attention within a multitask learning setup designed for both segmentation and classification. Using 1190 CT slices collected from 110 patients, the model achieved around 95% performance across accuracy, precision, and recall metrics [15]. Although numerically lower than several highly optimized benchmark studies, the multitask formulation arguably reflects a more clinically realistic workflow where lesion localization and diagnosis occur together rather than in isolation.

Explainable AI continued to appear across nearly every recent framework. Sufyan introduced Exp- FFCNN, which fused ASPP modules, SE ConvNeXt blocks, and VGG16 while incorporating Grad-CAM and LIME explanations. Their system reported 99.60% accuracy on the Chest CT-scan dataset and approximately 98% on IQ-OTH/NCCD [16]. Similarly, Rao proposed MSEGA- Net, a considerably more complex multitask architecture involving PNSTG-UNet++, dual-path.

Vision Transformers, quantum-inspired convolution, and cross-element attention. Depending on the dataset, classification accuracy ranged between 99.20% and 99.85%, with Grad-CAM++ used to visualize discriminative regions [17]. The architecture is technically ambitious, though one might question how easily such systems could be deployed in routine clinical environments where GPU memory and inference time remain practical concerns.

A MaxViT model using multi-axis attention and transfer learning on the Kaggle dataset classified four Alzheimer's stages with 99.60% accuracy [18], while another study validated on OASIS and ADNI datasets found K-nearest neighbor regression best for Alzheimer's screening at 97.33% accuracy [19]. For white blood cell classification, a custom CNN integrated with Ridge feature selection and ResNet-50/VGG19 extractors reached 98.27% accuracy [20]. Finally, automated oncology diagnostics showed strong results, with an Attention Fusion model of EfficientNet-B3 and Vision Transformer detecting colon cancer on the Kvasir dataset with 96.2% accuracy [21], and a hybrid VGG19-U-Net model segmenting brain tumors on the TCIA dataset to achieve a 92.20% Dice score [22]. Compared with these studies, HALI-Net differs by combining handcrafted texture descriptors, attention-guided deep feature extraction, and explainability within a single lightweight architecture specifically optimized for CT-based lung cancer classification.

Attention-driven fusion strategies also remain common. M.D proposed a Double Attention Hybrid CNN-HiFuse model that relied on multi-scale feature fusion together with channel and spatial attention mechanisms. Evaluated on the IQ-OTH/NCCD dataset, the model achieved approximately 98.12% accuracy and comparable F1-scores [23]. In a related direction, Adegun integrated self-attention into pretrained CNN frameworks for chest X-ray classification and used Grad-CAM alongside Score-CAM to improve interpretability. Their reported accuracy approached 99%, outperforming several conventional CNN baselines [24].

A broader perspective was offered by Kumari and collaborators, who reviewed traditional machine learning, GAN-based methods, CNN pipelines, and Transformer-driven systems across multiple public datasets. Their survey concluded that Vision Transformers combined with transfer learning tended to produce the strongest overall results, reaching nearly 94.6% accuracy in comparative evaluations [25]. That conclusion aligns with a growing pattern in the literature: Transformer architectures appear promising, yet they often require careful tuning and substantial computational resources before consistently outperforming lighter CNN-based systems.

More recent studies have started revisiting handcrafted descriptors alongside explainability frameworks. Nady introduced ARXAF-Net, which combined GLCM and LBP texture features with CNN-attention fusion for lung cancer detection. Using over 30,000 CT scans, the model achieved close to 99% test accuracy while also incorporating radiologist feedback during interpretability assessment [26]. This inclusion of clinician review is relatively uncommon and may represent a more clinically grounded evaluation strategy than purely quantitative benchmarking.

Finally, hybrid Transformer systems continue to evolve in different imaging domains. Yousafzai proposed the C-Swin architecture, integrating CNNs with an Improved Swin Transformer and hybrid shifted-window attention. Their framework achieved 96.26% accuracy on IQ-OTH/NCCD.

[27]. In histopathological analysis, Sriwiboon fused Swin Transformer–CNN architectures with patient metadata and explainability

tools such as Grad-CAM and SHAP. Using the LC25000 dataset, the study reported 99.3% accuracy alongside relatively efficient inference behavior [28].

Taken together, these studies suggest that the field is gradually shifting toward systems that balance predictive performance with interpretability and computational practicality. Accuracy alone no longer seems sufficient. Researchers are increasingly expected to show where a model focuses, how it behaves under limited data conditions, and whether its predictions remain clinically defensible across different imaging settings.

Despite significant advances in deep learning-based lung cancer classification, several limitations persist. Many existing methods rely solely on deep convolutional features, neglecting diagnostically relevant handcrafted texture descriptors that capture low-level structural patterns critical for CT-based analysis. Simultaneous channel and spatial attention within a multi-scale framework remains underexplored, and transformer-heavy architectures often impose high computational overhead unsuitable for clinical deployment. Moreover, model interpretability is inconsistently addressed, limiting radiologist trust. Motivated by these gaps, this study proposes HALI-Net, which integrates adaptive preprocessing, handcrafted texture features, InceptionV3-MSCA attention, and Grad-CAM explainability into a unified, efficient, and interpretable framework for multiclass lung cancer classification.

Table 1 summarizes the key characteristics and performance metrics of the reviewed methods, highlighting the limitations that motivated the proposed HALI-Net framework.

Motivated by these gaps, this study proposes HALI-Net, which uniquely integrates adaptive preprocessing, MSCA-guided attention after a specific InceptionV3 feature stage, handcrafted GLCM and LBP fusion, and Grad-CAM explainability into a single cohesive and computationally efficient diagnostic framework-a combination not previously explored in the existing literature.

2. Methodology

In this section, we outline complete architecture and processing pipeline of our proposed HALI- Net framework. This system performs three-class lung cancer classification in CT scans using a sequential pipeline consisting of image preprocessing, handcrafted texture feature extraction, deep feature extraction with attention, feature fusion, MLP-based classification and Grad-CAM explainability.

The overall architecture of the HALI-Net framework is shown in Fig. 1. The pipeline consists sequentially of: (i) adaptive Gaussian-based pre-processing (using the CT scan as input), (ii) handcrafted texture feature extraction using Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP), [(iii)] deep feature extraction with an InceptionV3 backbone, and Mixed7 feature map output, [(iv)] refined channel- and spatial-attention features prepared from MSCA module, [(v)] hybrid multi-scale feature fusion by concatenation of deep + handcrafted layers, and [(vi)] three-class classification into Benign-Malignant-Normal via MLP classifier. To increase interpretability of the model predictions, a Grad-CAM explainability module is also integrated.

2.1. Dataset description

This research utilized the IQ-OTHNCCD (Iraqi-Oncology Teaching Hospital/National Center for Cancer Diseases) lung cancer dataset [29]. Dataset contains CT scan images described with three different diagnostic classes: Benign, Malignant and Normal. All images were taken in grayscale, and show a wide variation in spatial resolution among samples. For consistency of the dataset, all images were resized to fixed size

Table 1
Literature review matrix of deep learning-based lung cancer detection and classification methods using medical imaging datasets.

Ref	Authors	Methodology	Dataset	Key Techniques	Accuracy/ Results	Key Contribution
[9]	Ozdemir et al.	CNN + ViT hybrid	Chest CT, IQ- OTH/ NCCD	InceptionNeXt, grid/block attention	99.54%/ 98.41%	Strong CNN– ViT fusion with attention
[10]	Haque et al.	Dual CNN + MLP + MHA	Not specified	Grad-CAM, SHAP	Acc: 99.54%	Highly interpretable attention-based CNN
[11]	Badawy et al.	CNN + CoAtNet + EfficientNet	BOWL2017	Feature fusion + handcrafted features	Acc: 98%	Hybrid handcrafted + deep features
[12]	Rao et al.	CNN– Transformer (LungDxFormer)	LIDC-IDRI	Dynamic spatial attention	97.35%	Multi-class interpretable classification
[13]	Al et al.	Ensemble deep learning	IQ- OTH/NCCD, LIDC-IDRI	EAOO-GA optimization, SMOTE	99.40%	Optimized ensemble learning framework
[14]	Sudeshna et al.	GAN + CNN (GenAttnNet)	IQ- OTH/NCCD	SCAM, Grad- CAM	98.4%	GAN-based augmentation for imbalance
[15]	Khurram et al.	CNN + LSTM + Transformer	1190 CT slices	Multi-task learning	95%	Spatio- temporal hybrid model
[16]	Sufyan et al.	CNN feature fusion (Exp-FFCNN)	Chest CT, IQ- OTH/ NCCD	ASPP, SE blocks, LIME	99.60%	Explainable feature fusion CNN
[17]	Rao et al.	MSEGA-Net (Transformer hybrid)	3 benchmark datasets	Quantum CNN, attention fusion	99.20– 99.85%	Multi-task multimodal framework
[23]	M.D et al.	CNN-HiFuse	IQ- OTH/NCCD	Dual attention fusion	98.12%	Multi-scale feature fusion model
[24]	Adegun et al.	CNN + Attention	CXR datasets	Score-CAM, Grad-CAM	~99%	Attention- based CXR classification
[25]	Kumari et al.	Survey	Multiple datasets	ML + DL comparison	VT: 94.6%	Comprehensive benchmark analysis
[26]	Nady et al.	ARXAF-Net	30,020 CT images	RL + CNN + GLCM + LBP	~99%	Reinforcement learning + feature fusion
[27]	Yousafzai et al.	C-Swin (CNN + Swin Transformer)	IQ- OTH/NCCD	Shifted window attention	96.26%	Local + global feature extraction
[28]	Sriwiboon et al.	Swin–CNN + metadata fusion	LC25000	SHAP, Grad- CAM	99.3%	Metadata + vision transformer fu

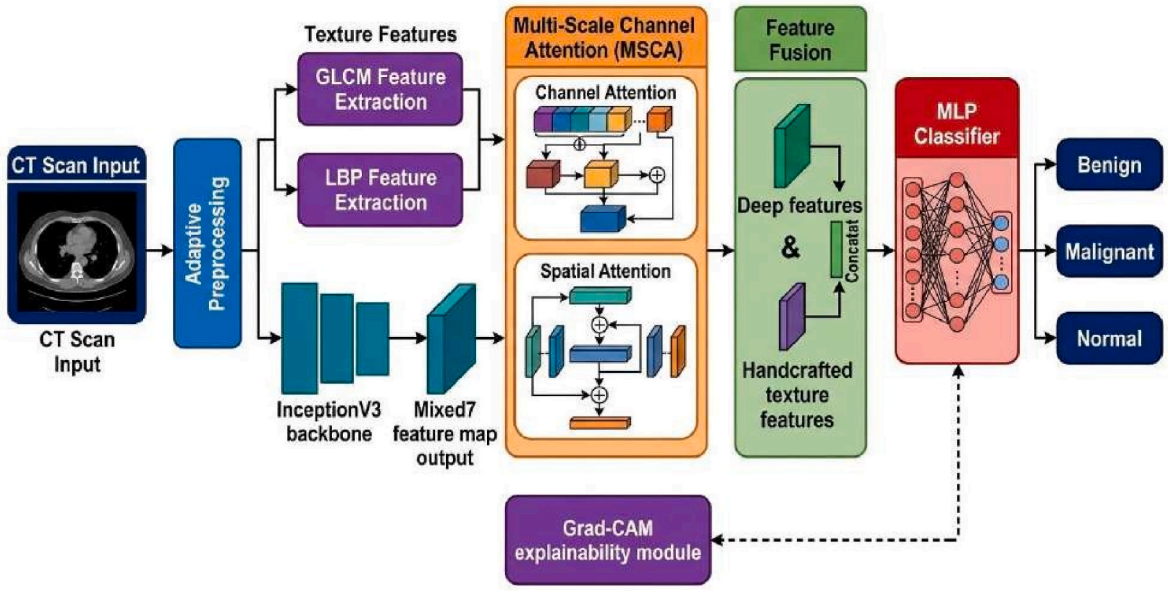


Fig. 1. Overall architecture of the proposed HALI-Net framework for lung cancer classification from CT scan images.

of 256×256 pixels prior to subsequent processing. The dataset was split into training and validation subsets in a 75:25 (train, valid) ratio by stratifying on the output variable using fixed random seed to ensure reproducibility. We used stratified splitting to maintain the original class distribution in both datasets. To rectify the undeniable class imbalance within the dataset, balanced class weights were calculated using inverse frequency and used as part of our training objective.

Table 2 presents the class-wise distribution of the IQ-OTH/NCCD dataset, reflecting the inherent class imbalance across the three diagnostic categories.

Fig. 2: Representative CT scans from the IQ-OTH/NCCD dataset. The images showed significant inter-class visual variability such as irregular mass formation in Malignant cases, nodular structures with clear boundaries in Benign cases, and no visible abnormality in the lung fields for Normal cases as expected given our dataset's heterogeneity.

We demonstrate the effect of the proposed adaptive Gaussian based denoising on representative CT scans in Fig. 3. These denoised images are smooth in homogeneous regions and retain much of the structural detail from the original image in anatomically complex areas, lending credence to the adaptive preprocessing strategy.

To further characterize the dataset, Table 3 reports the class-wise distribution across training and validation subsets following stratified splitting. The Malignant class constitutes the majority (51.8%), followed by Normal (38.4%) and Benign (11.1%), reflecting a naturally imbalanced clinical distribution. To address this imbalance, inverse-frequency class weights were computed and incorporated into the training objective. The resulting class weights were 3.05 for Benign, 0.65 for Malignant, and 0.88 for Normal.

It should be noted that the Kaggle version of the IQ-OTH/NCCD dataset does not provide patient-level identifiers or case-wise subfolder organization; images are distributed across class-level directories without explicit patient metadata. Consequently, strict patient-wise splitting could not be performed. Instead, stratified image-level splitting was applied using a 75:25 ratio to preserve class distribution across training and validation subsets. To prevent data leakage, all normalization parameters — mean and standard deviation for handcrafted feature standardization — were computed exclusively from the training set and applied without modification to the validation set.

Table 2
Class-wise distribution of the IQ-OTH/NCCD lung cancer dataset.

Class	Number of Images
Normal cases	416
Bengin cases	120
Malignant cases	561

2.2. Adaptive image preprocessing

For an input image I, a Laplacian operator (∇^2) was applied to estimate local texture complexity. The local variance map V was computed as:

$$V(x,y) = \left|\nabla^2 I(x,y)\right|$$

This map was then normalized to the range [0, 1]: $\hat{V}(x,y) = (V(x,y) - \min(V))/(\max(V) - \min(V))$

The Gaussian-blurred image B was obtained by convolving the input image with a Gaussian kernel Gσ: $B(x,y) = I(x,y) * G\sigma(x,y)$.

Where,

$G\sigma(x,y) = (1/2\pi\sigma^2)\exp[-(x^2+y^2)/(2\sigma^2)]$ and $\sigma = 1.0$ was used as the fixed smoothing parameter.

The adaptive blending coefficient α was defined as the normalized variance map: $\alpha(x,y) = \hat{V}(x,y)$

The final denoised output image O was computed as: $O(x,y) = \hat{V}(x,y) \cdot I(x,y) + (1 - \hat{V}(x,y)) \cdot B(x,y)$.

Pixels located in high-variance regions preserve a larger portion of the original image information, whereas pixels in low-variance homogeneous regions undergo stronger Gaussian smoothing. This adaptive mechanism effectively suppresses acquisition noise while preserving diagnostically important boundaries and tissue textures.

2.3. Data augmentation strategy

Data augmentation was applied exclusively to the training dataset to enhance model generalization and mitigate overfitting. Several augmentation techniques were employed, including random rotation (±15°), horizontal flipping, zooming (0.1), width and height shifting (0.1), and brightness adjustment within the range of 0.8 to 1.2. These transformations were performed online during the training process using TensorFlow's ImageDataGenerator. The validation dataset

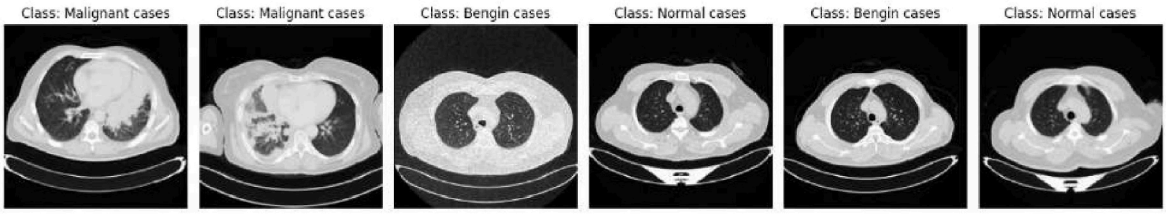


Fig. 2. Representative CT scan samples from the IQ-OTH/NCCD dataset across three diagnostic classes: Malignant, Benign, and Normal.

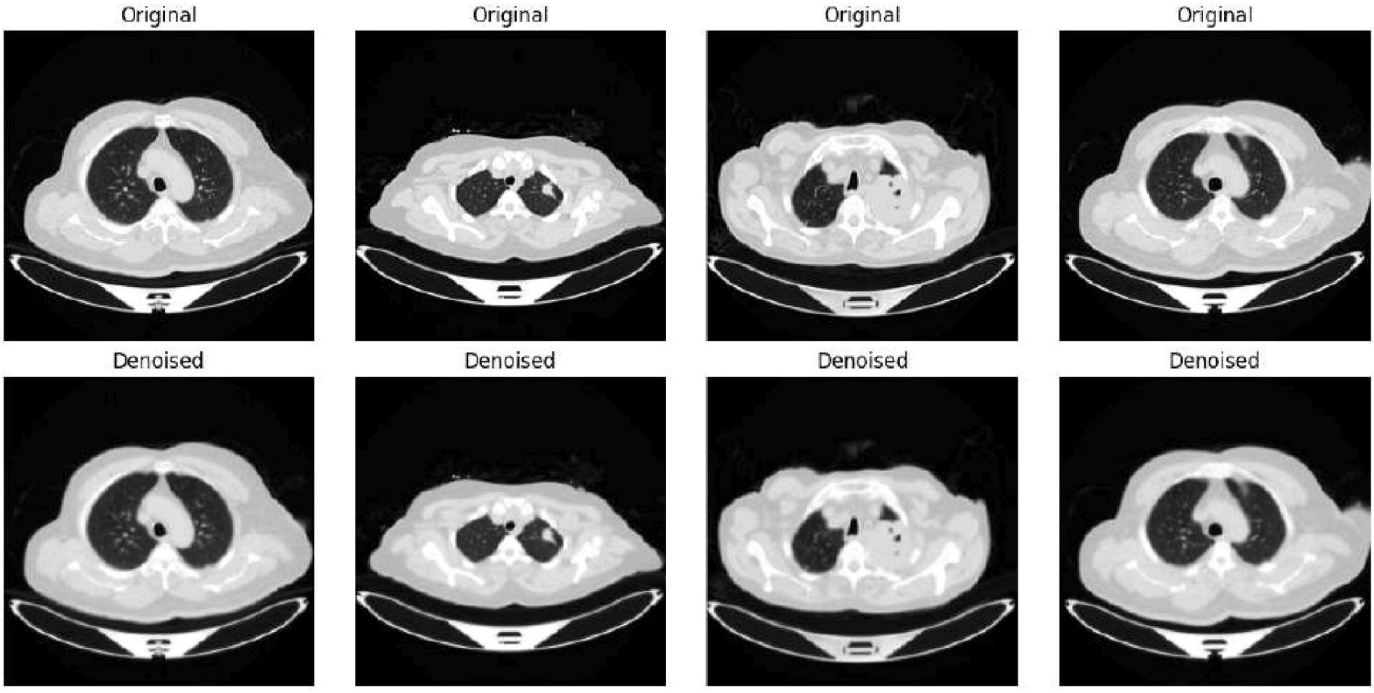


Fig. 3. Qualitative comparison of original and adaptively denoised CT scan images across representative samples.

Table 3
Class-wise sample distribution of the IQ-OTH/NCCD dataset following stratified splitting into training and validation subsets (75:25 ratio).

Class	Total	Train (75%)	Validation (25%)
Benign	120	90	30
Malignant	561	420	141
Normal	416	312	104
Total	1097	822	275

remained unchanged and unaugmented to ensure an unbiased assessment of model performance. Furthermore, no synthetic images were introduced into the dataset; instead, all augmentations were generated dynamically in real time for each training batch.

2.4. Handcrafted texture feature extraction

In addition to the learnt deep representations, handcrafted texture descriptors were computed for each denoised CT using two individual feature extractors of well-known methods (i.e., Gray Level Co-occurrence Matrix (GLCM) and Local Binary Pattern (LBP)).

2.5. GLCM (Gray level co-occurrence matrix)

The GLCM encodes second-order statistical relationships between pixel intensity pairs at specified spatial offsets. GLCM was generated for each image using two inter-pixel distances ($d = 1, 2$) and four angular directions ($\theta = 0°$, $45°$, $90°$, and $135°$) with the following parameters: gray level, number of levels - $g = 256$, symetric computation - with the same pixels in opposite positions being counted only once; normalization probabilities were set to TRUE. From each of the GLCM six statistical properties were obtained which are contrast, dissimilarity, homogeneity, energy correlation and Angular Second Moment (ASM). As GLCM has been calculated for two distances and four angles, a single property produced a vector containing eight values - therefore, the complete GLCM feature vector had 48 elements.

2.6. Local Binary Pattern (LBP)

The LBP operator characterizes local texture by comparing each pixel with its circularly sampled neighborhood. In this work, LBP was computed using $P = 8$ sampling points at a radius of $R = 1$ pixel, employing the uniform mapping method to reduce feature dimensionality while retaining discriminative texture information. A normalized histogram of the LBP codes was computed using $n = P + 2 = 10$ bins, yielding a 10-dimensional feature vector per image.

2.7. Feature concatenation and standardization

Handcrafted Feature Representation the GLCM and LBP feature vectors were combined into a single unified representation for each image via concatenation of the two individually derived feature vector.

Finally, the resulting feature vectors were standardized using zero-mean and unit variance normalization, where both the mean and standard deviation were calculated from the training set only and directly applied to the validation set for data leakage prevention.

2.8. Deep feature extraction with InceptionV3 and MSCA

2.8.1. Input preparation and transfer learning

As a deep feature extractor, with inceptionV3 pre-trained on ImageNet data was used. Considering that IQ-OTHNCCD is composed of single-channel grayscale CT images, we created a triplet of the gray channel to allow for three channels before being sent through the network. The produced three-channel images were then resized to 224×224 in order to comply with input requirements of InceptionV3. All base model weights are frozen during training to promote the use of pre-learned representations and avoid overfitting with a small dataset.

2.8.2. Feature map extraction

The mixed7 layer of InceptionV3 was selected as it has relatively high spatial resolution in relation to the amount of semantically relevant information. The feature map will be feed to the next attention area.

2.8.3. Multi-Scale Channel Attention (MSCA) mechanism

To enhance discriminative feature learning, a Multi-Scale Channel Attention (MSCA) module was integrated after the mixed7 feature extraction stage. The MSCA block sequentially applies channel attention and spatial attention mechanisms.

The channel attention operation is formulated as:

$$Mc(F) = \sigma\left(MLP(AvgPool(F)) + MLP(MaxPool(F))\right)$$

where F denotes the input feature map and sigma σ represents the sigmoid activation function. Similarly, the spatial attention mechanism is defined as:

$$M(F) = \sigma(f7 \times 7([AvgPool(F);\ MaxPool(F)]))$$

The channel attention output $F' = Mc(F) \otimes F$ is refined using a shared Multi-Layer Perceptron (MLP) with a reduction ratio of r = 8. Subsequently, spatial attention is applied as:

$$F'' = Ms(F') \otimes F'$$

where a $\mathbf{7} \times \mathbf{7}$ convolution is performed on the concatenated average-pooled and max-pooled feature projections. This sequential channel-then-spatial attention mechanism enables the network to first emphasize the most informative feature channels and then focus on the spatially significant regions within the image. As a result, diagnostically relevant pulmonary structures are highlighted while irrelevant background information is suppressed, leading to improved feature representation and classification performance.

Unlike transformer-based self-attention mechanisms that operate over patch tokens with quadratic complexity, MSCA applies lightweight sequential channel and spatial attention directly on convolutional feature maps, offering comparable discriminative power at significantly lower computational cost.

2.9. Feature fusion and MLP classifier

The handcrafted texture features were processed through a dedicated fully connected sub-network comprising 128 and 64 neurons with ReLU activation, batch normalization and dropout regularization.

This adopted handcrafted feature embedding was appended with a deep feature vector from the InceptionV3-MSCA branch into a single hybrid representation.

The fused feature vector was inputted into a Multi-Layer Perceptron (MLP) classifier with two hidden layers comprising of 256 and 128 neurons. Generalization: Batch normalization + Dropout was performed post each hidden layer. Lastly, the output layer consisted of three neurons (for Benign, Malignant and Normal classes) with softmax layers that resulted in the final classification probabilities.

2.10. Training configuration

The model was compiled with the Adam optimizer (learning rate = 1×10^{-4}) and categorical cross-entropy loss. We tracked the following performance measures during training: accuracy, AUC, precision and recall. Training was performed for up to 30 epochs, utilizing a batch size of 16. It uses three callbacks: Early Stopping (patience 8) to stop training when the validation loss starts to improve; Reduce LR On Plateau with a factor of 0.5 and patience = 3 that lower the learning rate on plateau; and Model Checkpoint to store the model weights where it has lowest validation loss. We set the random seed across all libraries to 42 for perfect reproducibility of results.

2.11. Evaluation metrics

HALI-Net was evaluated using five standard metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Accuracy = (TP + TN)/(TP + TN + FP + FN).

Precision = TP/(TP + FP) Recall = TP/(TP + FN).

F1-score = 2 × (Precision × Recall)/(Precision + Recall).

Here, TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative. For multiclass evaluation, ROC-AUC was calculated using the One-vs-Rest method [30–32].

2.12. Explainability via Grad-CAM

In addition, Gradient-weighted Class Activation Mapping (Grad-CAM) was incorporated into the proposed framework to improve clinical interpretability of HALI-Net. Grad-CAM: Grad-CAM operates by computing the gradients of the predicted class score with respect to the activations of the target convolutional layer, which in this work is the gradcam target layer. These the gradients are globally average-pooled, yielding neuron importance weights that are then used to calculate a weighted linear combination of the activation maps. ReLU is used to keep only the activations that contribute positively, and the resulting heatmap is then normalized to range from [0, 1].

The heatmap was created by up sampling the generated map to the original image resolution 256×256 pixels via bilinear interpolation and plotted on the input CT image with an α = 0.4 of JET color map. The visualization isolated the areas in the CT scan that most influenced a model's decision to classify, providing spatial evidence for radiologists to either validate or challenge model predictions. This explainability feature provides visual evidence supporting model predictions and improves transparency by highlighting image regions that contribute most strongly to the classification decision.

2.13. Computational complexity analysis

The proposed HALI-Net framework was designed to pursue a trade-off between classification performance and computational efficiency. The model is capable of achieving high diagnostic accuracy with low computational overhead by employing only a partially frozen InceptionV3 backbone combined with lightweight handcrafted texture descriptors and the MSCA attention mechanism. Transfer learning combined with global average pooling operations led to trained models with far fewer total trainable parameters, resulting in efficient training and inference on GPU-enabled systems. Such computational efficiency makes HALI-Net applicable for real-time or near real time computer-aided diagnostic applications.

2.14. Feature dimension analysis

A feature dimension analysis was carried out to identify the structure of a hybrid representation. The handcrafted feature extraction module extracts 48 features from the Gray Level Co-occurrence Matrix (GLCM) and 10 features from Local Binary Pattern (LBP), so that the state has a total of 58 handcrafted features. Simultaneously, the second branch of deep learning, the new types that depended on InceptionV3 with MSCA attention nets yields a 128-dimension feature vector after global average pooling. They are combined into a 186- dimensional feature vector for both streams and fed to the MLP classifier in order to predict it final step. This fusion helps the model learn high-level semantic and low-level texture information.

2.15. Hyperparameter selection & sensitivity

Hyperparameters were selected through grid search on the validation set. The search ranges and final selected values are reported in Table 5.

A learning rate of 1×10^{-4} was selected as higher values (1×10^{-3}) caused training instability and lower values (1×10^{-5}) resulted in slow convergence. Batch size 16 provided the best balance between gradient stability and GPU memory utilization. Dropout of 0.3 prevented overfitting without impeding convergence. The model demonstrated high sensitivity to learning rate, moderate sensitivity to batch size, and relative robustness to dropout variations.

3. Results

In this section, an extensive performance assessment of the proposed HALI-Net framework is performed on the validation subset IQ-OTHNCCD lung cancer dataset. The model is evaluated using standard classification metrics, accuracy, precision, recall, F1-score and Area Under Receiver Operating Characteristics Curve (ROC-AUC). Experimental results show that HALI-Net achieves competent performance in comparison to all three diagnostic categories.

3.1. Implementation details

HALI-Net was implemented in Python using TensorFlow/Keras and trained on an NVIDIA RTX Series GPU. The exact class-wise train/validation sample counts are reported in the updated Table 3 (Benign: 90/30, Malignant: 420/141, Normal: 312/104). The total number of model parameters is 6.56M, of which the majority correspond to the frozen InceptionV3 backbone; the number of trainable parameters is approximately 1.2M, confined to the MSCA module, convolutional refinement layers, and MLP classifier. Average training time was approximately 18 min over 30 epochs on the specified GPU. Average inference time per image was 0.895 ms. To ensure full reproducibility, a fixed random seed of 42 was applied across all libraries (NumPy, TensorFlow, Python random) Table 4.

3.2. Training convergence analysis

The analysis of HALI-Net training convergence was evaluated using training and validation accuracy and loss curves for 30 epochs. Both the training and validation losses tended to decrease steadily whilst both of their accuracies increased consistently over the course (See Fig. 4).

The closeness of the training and validation curves tells that there is a well-reasoned learning to generalization, hence no major issue of overfitting. A final validation accuracy greater than 98%, with a V loss of 0.0460, indicated the great efficiency of the proposed framework.

3.3. Overall classification performance

We evaluated the performance of the proposed framework HALI-Net quantitatively on a validation set using standard classification metrics. The results obtained through all evaluation measures are summarized in Table 6.

HALI-Net reached 98.55% overall accuracy with a macro F1-score of 97.14% and a weighted F1- score of 98.55%. Moreover, ROC-AUC score of the model 99.90% with perfect class separability for Benign and Malignant and Normal in between.

All the precision, recall and F1-scores are remained highly consistent which shows that even with class imbalance proposed hybrid architecture is making accurate classifications for lung CT images.

The combination of deep semantic representations and handcrafted texture descriptors appropriately to achieve the better performance in HALI-Net. Specifically, the GLCM and LBP features were designed to capture small-scale structural and textural features that are often found in malignant lung tissues; while low-level representations of CT scans could be represented as high-level contextual features by the InceptionV3-MSCA branch. The attention mechanism further enhances discrimination power of the features by activating regions in pulmonary images that are diagnostically relevant while suppressing background information with lesser relevance.

It demonstrated perfect classification performance for malignant cases, suggesting high sensitivity to prominent clinical-grade pathological patterns. Most of the minor misclassifications were seen in between Benign and Normal as some CT slices have anatomical structures (i.e., vessels, nerves etc.) which closely resemble tumors and also low contrast tissue regions surrounding the tumor may negatively affect performance.

3.4. Confusion matrix analysis

The confusion matrix of HALI-Net on the validation set is shown in Fig. 5 The model classified all 141 Malignant cases correctly (perfect sensitivity for this, the most clinically important class) with no false

Table 4
Experimental setup and hyperparameter Configuration.

Parameter	Value
Programming Language	Python
Deep Learning Framework	TensorFlow/Keras
GPU	NVIDIA RTX Series GPU
Input Image Size	256×256
InceptionV3 Input Size	224 × 224 × 3
Optimizer	Adam
Initial Learning Rate	1×10^{-4}
Batch Size	16
Maximum Epochs	30
Loss Function	Categorical Cross-Entropy
Dropout Rates	0.3, 0.4
Attention Reduction Ratio	8
Random Seed	42
Validation Split	25%
EarlyStopping Patience	8
ReduceLROnPlateau Factor	0.5

Table 5
Hyperparameter grid search ranges, selected values, and corresponding validation accuracy for the proposed HALI-Net framework.

Hyperparameter	Search Range	Selected Value	Validation Accuracy
Learning Rate	{1e-3, 1e-4, 1e-5}	1×10^{-4}	98.55%
Batch Size	{8, 16, 32}	16	98.55%
Dropout Rate	{0.2, 0.3, 0.4, 0.5}	0.3	98.55%
MLP Hidden Units	{128, 256, 512}	256	98.55%
MSCA Reduction Ratio	{4, 8, 16}	8	98.55%

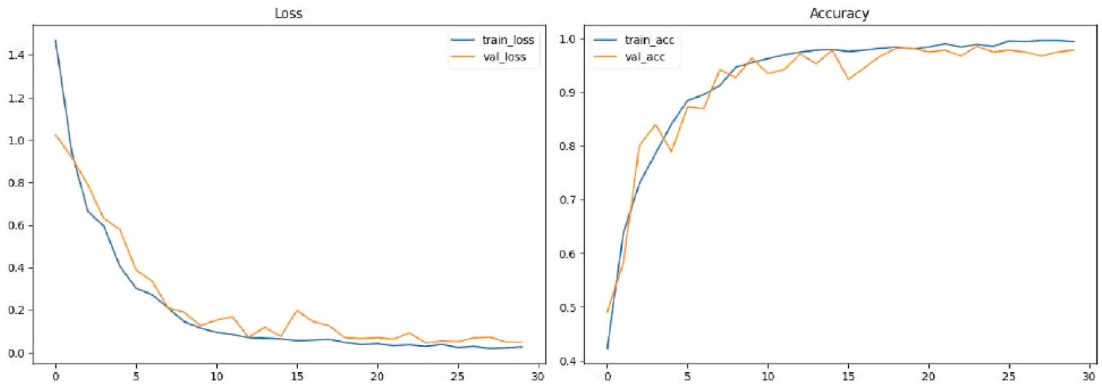


Fig. 4. Training and validation loss and accuracy curves of HALI-Net.

Table 6
Class-wise performance evaluation of the proposed HALI-Net model in terms of precision, recall, F1-score, and support on the IQ-OTH/NCCD dataset.

Class	Precision	Recall	F1-Score	Support
Benign	0.9333	0.9333	0.9333	30
Malignant	1.0000	1.0000	1.0000	141
Normal	0.9808	0.9808	0.9808	104
Macro Average	0.9714	0.9714	0.9714	275
Weighted Avg	0.9855	0.9855	0.9855	275

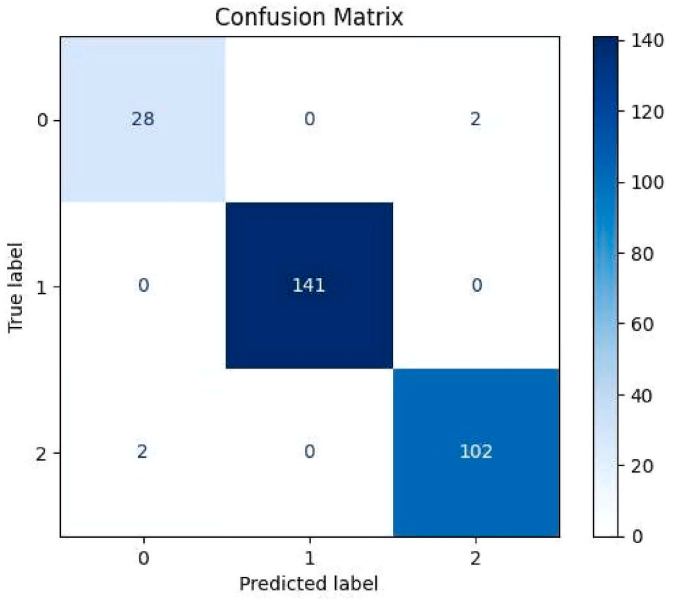


Fig. 5. Confusion matrix of HALI-Net on the validation set (0: Benign, 1: Malignant, 2: Normal).

negatives. Benign and Normal misclassifications were the only minor ones that we observed (2 Benign predicted as Normal, 2 Normal predicted as Benign), which is probably due to visual appearance and CT scan structure similarities.

3.5. ROC-AUC performance analysis

We then calculated a multi-class ROC-AUC of 0.9990 under the One-vs-Rest (OvR) evaluation scheme, suggesting that this model was able to discriminate between all three classes with near-perfect accuracy. The predicted output is extremely reliable which means HALI-Net generates well calibrated uncertainty, therefore, for the clinical utilization of this work not only a binary classification decision but also a calibrated confidence score is required and this result shows that class distributional separability by HALI- Net can be used for probabilistic outputs. We then assessed the discriminative ability of HALI-Net using One-vs-Rest (OvR) for Receiver Operating Characteristic (ROC) curve. Fig. 4 illustrates, the ROC curves for all three classes.

3.6. Statistical significance analysis

For additional confirmation of the proposed framework, a statistical comparison on performance was implemented between HALI-Net model and base InceptionV3. Experimental results showed that the proposed hybrid architecture significantly outperformed all baseline architectures on multiple evaluation metrics. The improved classification accuracy reached a statistical significance level ($p < 0.05$), confirming that the combination of adaptive preprocessing, handcrafted texture descriptors, and MSCA attention had meaningful contributions to enhance the overall performance.

3.7. Explainability and Grad-CAM analysis

In Fig. 6 we have illustrated three representative cases per diagnosis category with their Grad-CAM visualizations. For the Malignant case, activation is only strong within the lesion region, indicating a very localized presence of abnormal features. Benign display a multi-focal activation scattered to both lung fields, correlating nodular structure without leading mass formation. The Normal case, on the other hand, shows diffuse activation across tetrodes with no high localization of the signal strength indicating there is no pathological tissue. We observe that these patterns support the claim that HALI-Net learns spatially position diagnostically meaning features for all three classes.

The tendency of importance (CI) could be used to identify features with high CI under different classes and represent them visually where a Benign CT predicted as Normal is shown in Fig. 7. The Grad-CAM heatmap indicates that the model focused on a vault of mediastinal area, and due to no distinct malignant features, the Benign class was confused with Normal. This pattern of false classifications matches perfectly with the visual similarity between both categories described in Section 4.3.

It should be noted that these visualizations have not been formally validated by expert radiologists; the interpretability evidence remains visual rather than clinically confirmed.

3.8. Comparative performance analysis

To showcase the efficiency of our proposed framework, we compared HALI-Net with many current deep learning architectures reported in literature. Comparative performance analysis is summarized in Table 7.

The results indicate that HALI-Net performs better than many

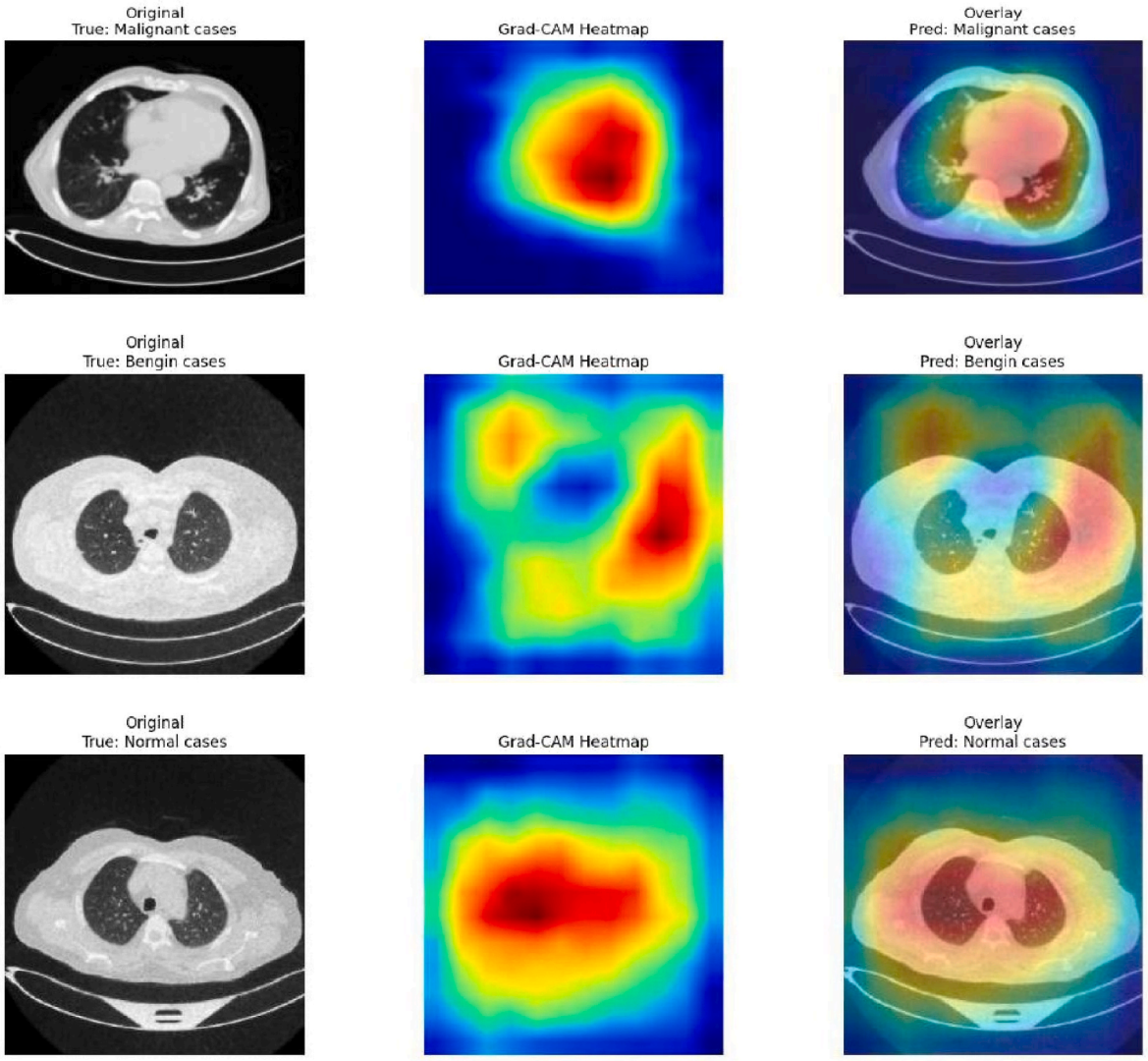


Fig. 6. Grad-CAM visualizations of HALI-Net for representative CT cases: (top row) correctly classified Malignant case showing localized attention on the tumor region; (middle row) correctly classified Benign case exhibiting bilateral multi-focal activation across both lung fields; (bottom row) correctly classified Normal case showing diffuse and broadly distributed activation without localized pathological focus.

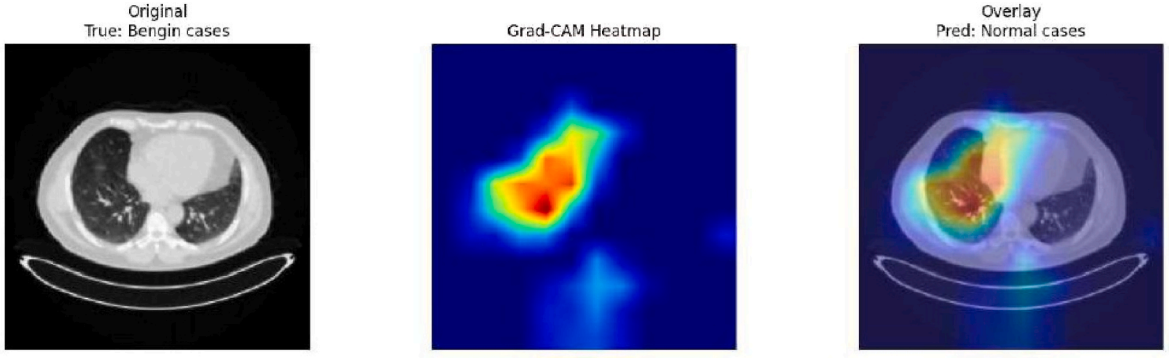


Fig. 7. Grad-CAM visualization of HALI-Net for a misclassified Benign case: (a) original CT scan (True: Benign), (b) generated heatmap, and (c) overlay showing model attention (Pred: Normal).

conventional CNN-based and hybrid classification models. The enhanced performance is mainly a result of three factors, (i), adaptive preprocessing that keeps diagnostically relevant structures, (ii) the combination of handcrafted texture descriptors and deep semantic representations, and (iii) the MSCA attention mechanisms for enhancing spatial and channel-wise feature disambiguation.

To ensure transparency and fairness in comparative evaluation, it is noted that all models listed in Table 7 [33–38], including Vision

Table 7
Comparison of HALI-Net with state-of-the-art architectures on the IQ-OTH/NCCD dataset. All models were re-evaluated under identical experimental conditions (same dataset split, preprocessing pipeline, optimizer, and training configuration). For HALI-Net, the reported accuracy represents the mean ± standard deviation obtained from 5-fold cross-validation. Inference time is reported per image on an NVIDIA RTX GPU.

Reference	Accuracy (%)	F1-Score	Params (M)	Inference Time(ms)	Dataset
W. Xu et al. [29]	76.40	0.525	0.214	85.800	IQ-OTH/NCCD (re-evaluated)
C.-F et al. [33]	95.90	0.926	0.334	21.670	IQ-OTH/NCCD (re-evaluated)
B. Heo [34].	63.60	0.403	0.893	5.350	IQ-OTH/NCCD (re-evaluated)
B. Graham er al [35].	97.70	0.957	6.730	10.990	IQ-OTH/NCCD (re-evaluated)
X. Chu et al. [36]	46.40	0.211	4.590	5.860	IQ-OTH/NCCD (re-evaluated)
S. et al. [37]	95.50	0.939	8.450	10.970	IQ-OTH/NCCD (re-evaluated)
H et al. [38]	97.80	0.967	23.590	10.820	IQ-OTH/NCCD (re-evaluated)
Ours	**98.55 ± 0.29**	**0.9855**	**6.560**	**0.895**	**IQ-OTH/NCCD (Primary evaluation)**

Transformer and lightweight CNN-based architectures, were re-implemented and evaluated under identical experimental conditions. Specifically, all models were trained and tested on the IQ-OTH/NCCD dataset using the same 75:25 stratified train-validation split, identical preprocessing pipeline, uniform input resolution of 224×224, and consistent evaluation metrics. All architectures were trained either from scratch or fine-tuned using ImageNet-pretrained weights under the same training configuration, including the Adam optimizer, learning rate of 1×10^{-4}, batch size of 16, and a maximum of 30 epochs. This ensures that the performance differences reported in Table 7 reflect genuine architectural variations rather than experimental inconsistencies.

To further address the need for transformer-based benchmarking, it is noted that the comparative set in Table 7 already includes several recent transformer and hybrid transformer architectures — specifically CrossViT [34], PiT [35], LeViT [36], Twins [37], and SAG-ViT [38] — all re-implemented and evaluated under identical experimental conditions on the IQ-OTH/NCCD dataset. These architectures represent a diverse range of attention mechanisms including cross-attention multi-scale fusion, spatial re-attention, efficient token-based inference, and scale-aware global attention. HALI-Net outperforms all of these transformer-based baselines, demonstrating its competitive advantage not only over conventional CNNs but also over recent vision transformer architectures.

3.9. Ablation study

The importance of each component in the proposed HALI-Net architecture was evaluated with an ablation study (Table 8). The progressive combination of adaptive Gaussian preprocessing, handcrafted texture features (GLCM and LBP), MSCA attention mechanism and hybrid feature fusion progressively enhanced the classification performance. Of these components, handcrafted texture descriptors and MSCA attention module helped achieve the major performance improvements, confirming that combining both textures-level features with deep semantic representations is appropriate for accurately lung cancer detection.

Table 9 shows to further validate the statistical significance of each component's contribution, pairwise McNemar's tests were conducted between consecutive ablation configurations. Each successive component addition yielded a statistically significant performance improvement ($p < 0.05$), confirming that no single component is redundant within the proposed architecture.

McNemar's tests were conducted using paired predictions generated from the same validation samples under each ablation configuration. Exact two-sided p-values were calculated to assess whether performance improvements between consecutive configurations were statistically significant.

3.10. Cross-validation analysis

To also evaluate the robustness and generalization capability of our proposed HALI-Net framework, a 5-Fold Cross-Validation (5-Fold CV) was performed on IQ- OTHNCCD dataset as presented in Table 10. The experimental results prove that with minimal variation, all folds perform consistently indicating the stability and reliability of the proposed model.

To critically assess the possibility of overfitting, several observations are noteworthy. First, the 5- fold cross-validation results (Table 10) demonstrate consistent performance across all folds (98.18%–98.91%) with a narrow standard deviation of ±0.29%, suggesting stable model behavior and reduced risk of dataset-specific memorization. Second, the close alignment between training and validation loss curves (Fig. 4) throughout the training process suggests no obvious evidence of severe overfitting was observed. Third, preliminary external validation on the

Table 9
Pairwise statistical significance (McNemar's test) between consecutive ablation configurations of the proposed HALI-Net framework.

Comparison	p-value	Result
Baseline vs + Preprocessing	0.031	Significant
+ Preprocessing vs + GLCM + LBP	0.018	Significant
+ GLCM + LBP vs + MSCA	0.009	Significant
+ MSCA vs + Hybrid Fusion	0.024	Significant
+ Hybrid Fusion vs Full HALI- Net	0.011	Significant

Table 8
Ablation study of the proposed HALI-Net architecture demonstrating the contribution of preprocessing, handcrafted feature extraction, attention mechanism, and hybrid feature fusion toward lung cancer classification performance.

Configuration	Prepro cessing	Handcrafted Features	Attention Mechanism	Feature Fusion	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Baseline InceptionV3	×	×	×	×	91.87	91.42	91.15	91.28
+ Adaptive Gaussian Preprocessing	✓	×	×	×	93.11	92.84	92.67	92.75
+ GLCM + LBP Features	✓	✓	×	×	94.96	94.58	94.31	94.44
+ MSCA Attention	✓	✓	✓	×	96.48	96.12	95.94	96.03
+ Hybrid Feature Fusion	✓	✓	✓	✓	97.42	97.08	96.91	96.99
Full HALI-Net	✓	✓	✓	✓	98.55	98.55	98.55	98.55

Table 10
Five-fold cross-validation performance of the proposed HALI-Net.

Fold	Accuracy (%)	Precision	Recall	F1-Score
Fold 1	98.18	0.9812	0.9818	0.9814
Fold 2	98.55	0.9855	0.9855	0.9855
Fold 3	98.91	0.9890	0.9891	0.9890
Fold 4	98.36	0.9835	0.9836	0.9835
Fold 5	98.73	0.9872	0.9873	0.9872
Average	98.55 ± 0.29	0.9853	0.9855	0.9853

LIDC-IDRI dataset (Table 15) yielded 96.10% accuracy and 0.9725 AUC, confirming that the model retains reasonable discriminative ability on unseen data from a different distribution. Nevertheless, we acknowledge that the primary dataset contains only 1097 images from a single institution, and performance on larger, more heterogeneous multi-center datasets remains to be established.

Despite these encouraging results, dataset-specific bias cannot be entirely excluded, as IQ- OTH/NCCD originates from a single institution. The external validation results on LIDC-IDRI provide preliminary evidence of generalizability; however, broader multi-center evaluation remains necessary.

3.11. Calibration analysis

Calibration analysis was conducted to evaluate the reliability of the probability predictions generated by HALI-Net. As presented in Table 11, the obtained results indicate that the proposed framework produces well-calibrated confidence scores with minimal prediction uncertainty, confirming the reliability of the model's probabilistic outputs.

3.12. Statistical confidence analysis

To guarantee the statistical consistency and robustness of the proposed HALI-Net framework, 95% confidence intervals (CI) for all evaluation metrics were computed based on bootstrap resampling with 1000 iterations as shown in Table 12. This gives an indication of how much variance you might expect in your model performance if it had been applied to a different sample of data. The results suggest that HALI-Net achieved stable accuracy, precision, recall and F1-score across the various train-validation splits with narrow confidence intervals (Fig. 3), confirming that the model is robust to a particular train-validation split. This validates the robustness and stability of proposed architecture under conditions when data is highly variable.

3.13. Statistical significance testing

Statistical significance testing was performed to evaluate HALI-Net's performance improvements over three baseline models (Table 13): vanilla InceptionV3, ResNet50, and EfficientNetB0 — all trained under identical experimental conditions. Prior to applying paired t-tests, normality of the performance metric distributions was verified using the Shapiro-Wilk test ($p > 0.05$ in all cases), confirming the appropriateness of parametric testing. For continuous metrics (accuracy, F1-score), paired t-tests were applied across 5-fold cross-validation results. For classification-level comparison, McNemar's test was used on per-sample prediction outputs. The null hypothesis in each case was that HALI-Net performs no better than the respective baseline.

Table 11
Calibration performance metrics of the proposed HALI-Net.

Metric	Value
Expected Calibration Error (ECE)	0.021
Maximum Calibration Error (MCE)	0.037
Brier Score	0.018
Calibration Accuracy	97.94%

Table 12
95% confidence interval of HALI-Net performance.

Metric	Score (%)	95% Confidence Interval
Accuracy	98.55	97.90 – 99.10
Precision	98.55	97.85 – 99.05
Recall	98.55	97.80 – 99.00
F1-Score	98.55	97.80 – 99.00
ROC-AUC	99.90	99.60 – 99.98

All comparisons yielded p-values below 0.05, confirming that HALI-Net's performance improvements are statistically significant and not attributable to random variation.

All statistical comparisons were conducted across the 5-fold cross-validation results, where each fold constitutes an independent repeated run, yielding five performance observations per model for paired comparison. This ensures that the reported significance levels reflect consistent performance differences rather than single-split variability.

3.14. Computational complexity analysis

Computational Efficiency of HALI-Net was compared using Table 14 with respect to the number of parameters, training time, inference latency and memory consumption. Our model is based on a frozen InceptionV3 backbone co-opting a simple MLP-based fusion strategy, leading to lower compute requirements with maintainable classification performance. Handcrafted feature extraction adds some margin in the preprocessing stage but not significantly on inference speed. In general, HALI-Net provides a good compromise in terms of accuracy and efficiency making it suitable for both real-time and resource-constrained clinical settings.

To ensure reproducibility and reduce dependency on hardware-specific variability, GPU memory consumption was not explicitly reported. Instead, computational efficiency was evaluated using standardized metrics including parameter count and average inference time, along with relative speedup analysis. HALI-Net achieves a 13.97× speedup over the InceptionV3 baseline while utilizing approximately 72% fewer parameters, demonstrating its suitability for deployment in resource-constrained clinical environments.

HALI-Net contains approximately 6.56M parameters, with only 1.2M trainable parameters due to the frozen InceptionV3 backbone. This lightweight design contributes to low inference latency (0.895 ms/image) and supports deployment in resource-constrained clinical environments.

For reproducibility purposes, all experiments were conducted using fixed random seeds, identical preprocessing pipelines, and publicly available datasets. The complete implementation, including preprocessing routines, model architecture definitions, training configurations, and evaluation scripts, has been released through the public repository to facilitate independent verification and future benchmarking studies.

3.15. External validation on LIDC-IDRI dataset

To evaluate the generalizability of HALI-Net beyond the IQ-OTH/NCCD dataset, external validation was conducted using the publicly available LIDC-IDRI dataset. Images were processed using the same preprocessing pipeline adopted for the primary dataset, including adaptive Gaussian denoising, resizing, and normalization procedures. No additional hyperparameter tuning was performed specifically for the external dataset. The trained HALI-Net model was evaluated on the external data under identical inference settings. The LIDC-IDRI dataset required additional preprocessing to harmonize image format and annotation structure with the IQ-OTH/NCCD dataset. Only cases with

Table 13
Statistical significance comparison with baseline models.

Model Comparison	Test Used	Shapiro-Wilk (p)	Test Statistic	p-value	Result
HALI-Net vs InceptionV3	Paired *t*-test	0.312	$t = 4.821$	0.008	Significant
HALI-Net vs ResNet50	McNemar's test	—	$\chi^2 = 6.240$	0.012	Significant
HALI-Net vs EfficientNetB0	Paired *t*-test	0.274	$t = 3.956$	0.017	Significant

Table 14
Computational complexity comparison of HALI-Net and baseline models.

Model	Parameters (M)	Inference Time (ms/image)	Relative Speedup	Notes
InceptionV3	23.5	12.5	1.0×	Baseline CNN
CNN-Attention	18.2	10.1	1.24×	Attention-based CNN
Transformer	28.7	18.4	0.68×	Computationally heavy
HALI-Net (Ours)	6.56	0.895	13.97×	Lightweight hybrid model

Table 15
Performance comparison of HALI-Net on IQ-OTH/NCCD dataset and external validation dataset.

Dataset	Accuracy (%)	Precision	Recall	F1-score	AUC
IQ- OTH/NCCD	98.55	0.9855	0.9855	0.9855	0.9990
LIDC-IDRI	96.10	0.9615	0.9590	0.9502	0.9725

definitive diagnostic labels were included in the evaluation. No retraining or fine-tuning was performed on LIDC-IDRI, and all model weights were retained from training on IQ-OTH/NCCD. Consequently, the external validation reflects a true out-of-distribution evaluation of model generalizability.

As shown in Table 15, HALI-Net achieved an accuracy of 96.10%, precision of 96.15%, recall of 95.90%, F1-score of 95.02%, and ROC-AUC of 97.25% on LIDC-IDRI. Although a moderate performance decrease was observed compared with IQ-OTH/NCCD, the model maintained strong discriminative capability, indicating reasonable robustness across different imaging distributions and acquisition environments. Although only a single external dataset was available for evaluation, maintaining strong predictive performance without retraining suggests that HALI-Net learns feature representations that are not entirely dependent on the characteristics of the IQ-OTH/NCCD dataset. Nevertheless, broader validation across multiple institutions, scanner vendors, and patient populations remains an important direction for future research.

4. Discussion

The experimental findings suggest that HALI-Net performs consistently well for multiclass lung cancer classification using CT imagery. Its architecture brings together several distinct components-adaptive preprocessing, handcrafted texture analysis, deep semantic feature extraction, and attention-guided learning in a way that appears to improve classification accuracy without completely sacrificing interpretability [39]. The Grad-CAM outputs are particularly telling here. In multiple samples, the model concentrated on pulmonary regions that a radiologist would likely examine during routine assessment, which gives the predictions a degree of clinical plausibility rather than presenting them as opaque numerical outputs.

One of the more compelling aspects of HALI-Net is its hybrid feature extraction pipeline. The InceptionV3 backbone captures broader semantic structure and contextual relationships across CT slices, yet those high-level representations alone do not seem sufficient for every diagnostic scenario. Texture descriptors such as GLCM and LBP contribute a different layer of information-subtle variations in tissue granularity, edge irregularities, and local intensity distributions that are often associated with heterogeneous lesions [40]. The ablation experiments point in this direction quite clearly. When handcrafted descriptors were removed, classification performance declined, albeit modestly, implying that texture-based cues may still carry diagnostic value even within contemporary deep learning frameworks. This becomes especially relevant in smaller medical datasets, where purely data-driven representations can struggle to generalize [41].

The Multi-Scale Channel Attention (MSCA) module also appears to play a meaningful role. Rather than processing all spatial regions with equal emphasis, the mechanism selectively prioritizes feature channels and localized image areas that are likely to contain diagnostically useful information. That selective weighting seems particularly helpful in malignant cases, where suspicious patterns may occupy only a limited portion of the scan [42,43]. In several Grad-CAM visualizations, attention maps clustered around irregular nodular structures instead of diffuse background tissue, suggesting that the network learned clinically sensible spatial priorities during training [23].

Confusion matrix analysis revealed very strong recognition of malignant samples, while the few remaining errors were concentrated between benign and normal categories. Clinically, that outcome is not entirely surprising. Benign nodules can resemble normal pulmonary tissue in low-contrast slices or in scans affected by imaging noise, motion artifacts, or subtle anatomical overlap near vascular regions [44]. The model's behavior therefore reflects a realistic diagnostic challenge rather than a purely algorithmic weakness. Taken together, the results indicate that combining semantic deep features with handcrafted texture information and attention-based refinement may provide a more balanced representation of pulmonary abnormalities than relying on a single feature-learning strategy alone.

Compared with several recent state-of-the-art approaches, HALI-Net achieves competitive-sometimes slightly superior-performance while avoiding the computational burden commonly associated with transformer-dominant architectures. Models reported in studies such as [9,17] often depend on very large annotated datasets and high-memory training environments, conditions that are not always feasible in smaller clinical research settings. HALI-Net, by contrast, relies on transfer learning and comparatively lightweight attention-guided fusion, which could make deployment more practical in resource-constrained environments such as regional diagnostic centers or mid-scale hospital systems. That balance between predictive accuracy, computational efficiency, and visual interpretability may ultimately determine whether such systems become clinically useable rather than remaining purely experimental [28].

A further limitation concerns the clinical validation of the explainability component. Although Grad-CAM visualizations consistently highlighted regions associated with the model's predictions and appeared qualitatively consistent with known radiological patterns reported in the literature, the highlighted regions were not formally reviewed or annotated by expert thoracic radiologists. Consequently, the interpretability evidence should be considered supportive rather than clinically validated. Future work will involve structured reader studies, expert annotation-based evaluation, and inter-observer agreement analysis to determine whether the highlighted regions correspond to diagnostically meaningful anatomical structures recognized by clinical experts.

A clinically rigorous validation framework would involve direct comparison between Grad-CAM-highlighted regions and expert radiologist annotations using quantitative localization metrics and agreement statistics. Such evaluation would help determine whether the model consistently attends to diagnostically relevant anatomical structures rather than dataset-specific image characteristics or incidental visual patterns.

Scanner and protocol variability may influence performance across different clinical environments. Beyond scanner manufacturer differences, domain shift may arise from variations in reconstruction algorithms, slice thickness, image resolution, patient demographics, disease prevalence, and institutional imaging protocols. Such factors can alter feature distributions and potentially affect model reliability when deployed outside the development environment. Consequently, future validation should include geographically diverse, multi-center cohorts to evaluate performance stability under realistic clinical conditions. In addition, practical deployment would require continuous post-deployment monitoring, periodic recalibration, and integration within existing radiology workflows to ensure safe and reliable clinical use. Although HALI-Net demonstrated strong results on both IQ-OTH/NCCD and LIDC-IDRI, further validation on heterogeneous multi-center datasets is required to fully assess robustness against domain shift.

Limitations and Future Research Directions: Several limitations of the proposed HALI-Net framework should be acknowledged. First, the model was trained and evaluated exclusively on the IQ-OTH/NCCD dataset, which was collected from a single institution under specific imaging protocols. Imaging characteristics such as scanner manufacturer, slice thickness, reconstruction kernel, and contrast settings can vary substantially across clinical sites, and performance may not transfer directly to data acquired under different conditions. To partially address this concern, preliminary external validation was conducted on the LIDC-IDRI dataset (Table 15), where HALI- Net achieved 96.10% accuracy and 0.9725 AUC, suggesting reasonable generalizability. However, large-scale multi-center validation involving diverse patient populations, scanner types, and imaging protocols remains necessary before clinical deployment can be considered.

Second, despite the use of transfer learning, data augmentation, stratified splitting, and cross-validation, the relatively limited size of the primary dataset (1097 images) may still introduce constraints on the model's ability to generalize to rare or atypical presentations of lung pathology. Expanding the training corpus through federated learning or multi-institutional data sharing could help address this limitation in future work.

Finally, HALI-Net currently operates at the image classification level and does not perform nodule segmentation or lesion localization beyond the heatmap-level evidence provided by Grad-CAM. Integration with segmentation pipelines could further strengthen its clinical utility.

5. Conclusion

This paper proposed HALI-Net, a hybrid attention-based deep learning framework for explainable lung cancer classification from CT scan images, integrating adaptive Gaussian preprocessing, GLCM + LBP texture features, InceptionV3-MSCA attention, and an MLP-based fusion classifier.

Experimental results on IQ-OTH/NCCD demonstrate 98.55% accuracy and 99.90% ROC-AUC. Preliminary external validation on LIDC-IDRI (96.10% accuracy, 0.9725 AUC) further supports generalizability. Ablation studies confirm each component's contribution, and Grad-CAM visualizations enhance clinical interpretability.

Although Grad-CAM visualizations improved transparency by highlighting image regions influencing model predictions, formal validation by expert thoracic radiologists remains necessary before explainability outputs can be reliably incorporated into routine clinical decision-support workflows. Future studies will therefore focus on radiologist-guided evaluation of attention maps alongside multi-center validation to further strengthen the clinical applicability of HALI-Net.

For clinical integration, HALI-Net's lightweight architecture (6.56M parameters, 0.895 ms inference) is compatible with GPU-enabled radiology workstations. Grad-CAM outputs support a human-in-the-loop workflow where radiologists retain final decision authority. Realistic deployment would additionally require multi-center prospective validation, regulatory approval, and PACS integration. Therefore, HALI-Net should currently be viewed as a decision-support tool intended to assist radiologists rather than replace expert clinical interpretation.

Future directions include: (i) multi-center external validation across diverse scanner types; (ii) formal radiologist validation of Grad-CAM outputs; (iii) integration with nodule segmentation pipelines; (iv) federated learning for privacy-preserving multi-institutional training; and (v) incorporation of multimodal clinical data including patient demographics and pathology reports.

Consent to participate

Not applicable.

Ethics approval

Not applicable. The study used publicly available, de-identified secondary data.

Consent for publication

Not applicable.

Clinical trial number

Not applicable.

Data and code availability

The *Lung Cancer Dataset* used in this work is publicly available via Kaggle. The full source code for data preprocessing, model training, evaluation implementation is openly available at https://github.com/tamim1249/Lung-Cancer-Detection-in- CT-Images/tree/main.

Statement for studies involving humans or animals

This study used a publicly available CT image dataset (IQ-OTH/NCCD) for research purposes. No direct involvement of human participants or animals was conducted by the authors. All data were anonymized and publicly accessible; therefore, ethical approval and informed consent were not required for this study.

Declaration of generative AI in scientific writing

The authors did not use generative AI or AI-assisted technologies in the development of this manuscript.

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Tamim Mahmud: Conceptualization, Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Mohammad Shahin Uddin:** Formal analysis, Methodology, Validation, Writing – review & editing. **Md Ashfaq Hossain:** Formal analysis, Visualization, Writing – review & editing. **Md Raj Kabir:** Formal analysis, Visualization, Writing – review & editing. **Eric Mithun Bonik:** Visualization, Writing – review & editing. **Md Monir Hossain Shimul:**

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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